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PALEOMAGNETISM OF TERTIARY ROCKS FROM THE OGA PENINSULA AND THE ROTATION OF NORTHEAST JAPAN

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Abstract. We have made a paleomagnetic study of Tertiary rocks from the Oga Peninsula, northern Honshu Island. Volcanic and sedimentary rocks were sampled from 26 sites in the peninsula. The ages of the rocks range from 62 Ma to the present. The results indicate a counterclockwise rotation of Northeast Japan with respect to eastern Asia around a pole at 58°N, 149°E, between about 22 Ma and 15 Ma. The amount of the rotation is about 20°. Before the rotation, Northeast Japan was situated along the east coast of the Asian continent.

INTRODUCTION

More than two decades ago, Kawai et al. [1961] demonstrated the bending of Honshu Island on the basis of the comparison of the paleomagnetic field directions from Northeast and Southwest Japan. Recent paleomagnetic studies in Japan have aimed at determining the mode and the timing of the bending and at relating the bending to the opening of the Sea of Japan. For Southwest Japan, Otofuji and Matsuda [1983] performed paleomagnetic investigations on the acidic rocks exposed on the Sea of Japan side of Southwest Japan and concluded that the clockwise rotation of that region took place from 28 to 12 Ma. The timing of the rotation was narrowed to around 15 Ma by Otofuji and Matsuda [1984].

For Northeast Japan, Tosha [1983] and Otofuji et al. [1985] recently made paleomagnetic studies. Otofuji et al. [1985] collected samples from a wide area of Northeast Japan, and their results indicate that the region north of the Tanakura Tectonic Line (see Figure 1a) behaved as one block, at least during the Tertiary period. However, the timing of the rotation of Northeast Japan is not clear from their data, since the relative ages of the rocks are unknown, partly because of the wide spread of the sampling sites. On the other hand, Tosha [1983] collected samples from the Oga Peninsula, where the sequence of Tertiary rocks is well preserved. Therefore his data are suitable for determining the chronologic progression of the movement of Northeast Japan. Recently, Celaya and McCabe [1987] and Moreau et al. [1987] used Tosha's [1983] data and interpreted the tectonic evolution of Japan. These workers discussed the opening of the Sea of Japan by using the temporal declination changes in Japan, and later workers argued the remagnetization of pre-Tertiary rocks. Since the thesis by Tosha [1983] is not widely known and the above authors' interpretation of his data is quite different from his own interpretation, we here present the original data and results of recent studies, and discuss the tectonic implications.

Some geological and geophysical studies suggest the tectonic continuity between the northeastern part of Honshu Island and the western part of Hokkaido Island in Cretaceous time [Takahashi, 1983; Takigami, 1983; Segawa and Furuta, 1978]. In the present study we assume the area north of the Tanakura Tectonic Line and the western part of Hokkaido to be one tectonic block since the Early Cretaceous, and we call the block Northeast Japan.

GEOLOGICAL SETTING, AND SAMPLING

In Northeast Japan, Tertiary and Quaternary deposits and volcanic rocks are widely exposed except in the Kitakami and Abukuma areas. Most of these rocks had been documented as Neogene to Quaternary in age. How-

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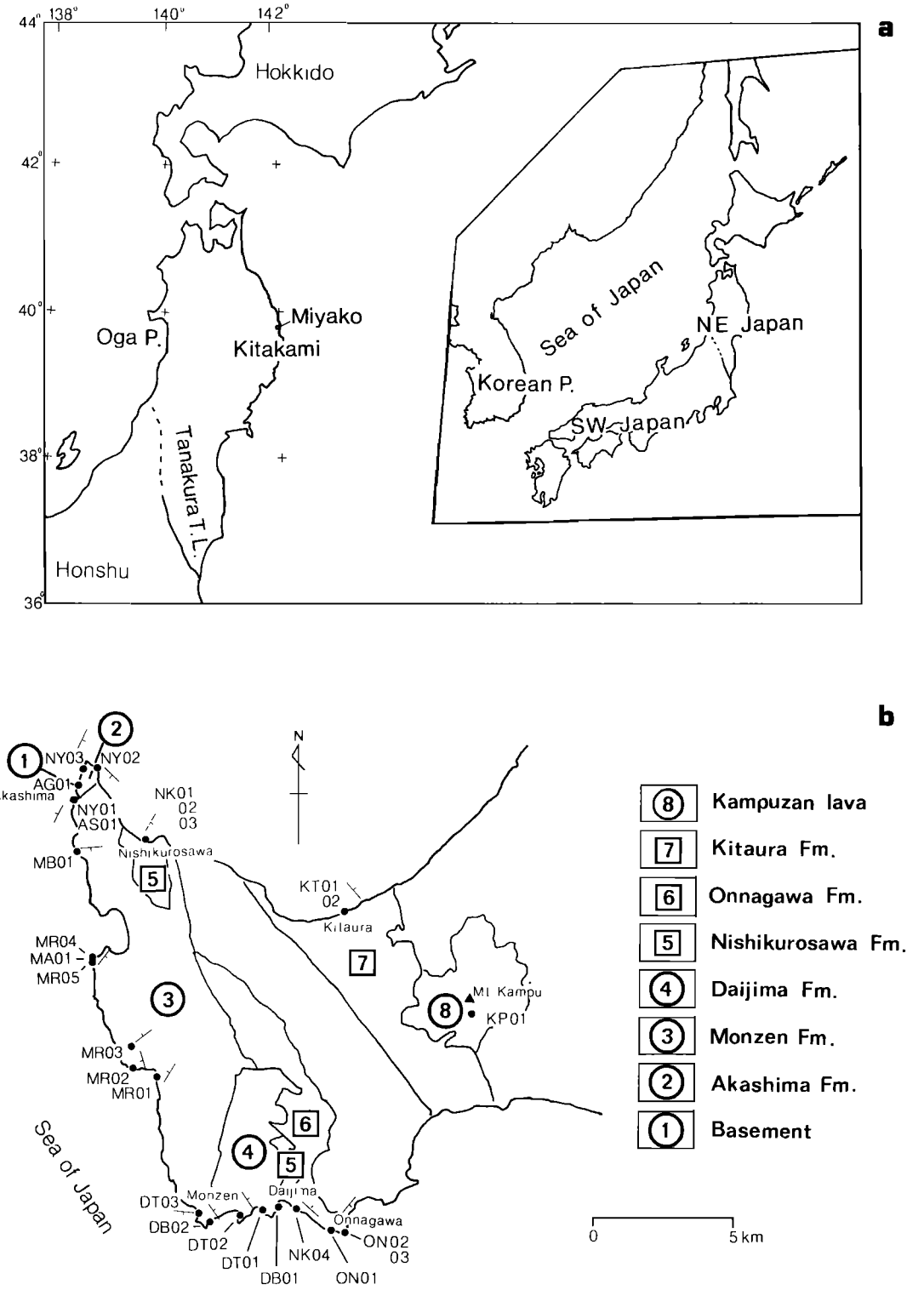


Fig. 1. (a) Location map of the area investigated in this study. The Oga Peninsula faces the Sea of Japan at about 40° north latitude. (b) Simplified geological map of the Oga Peninsula, after Takayasu and Matoba [1976]. Sampling sites and the directions of strike and dip at the sites are indicated. Basement crops out at a restricted area around the site AG01 in the Akashima Formation.

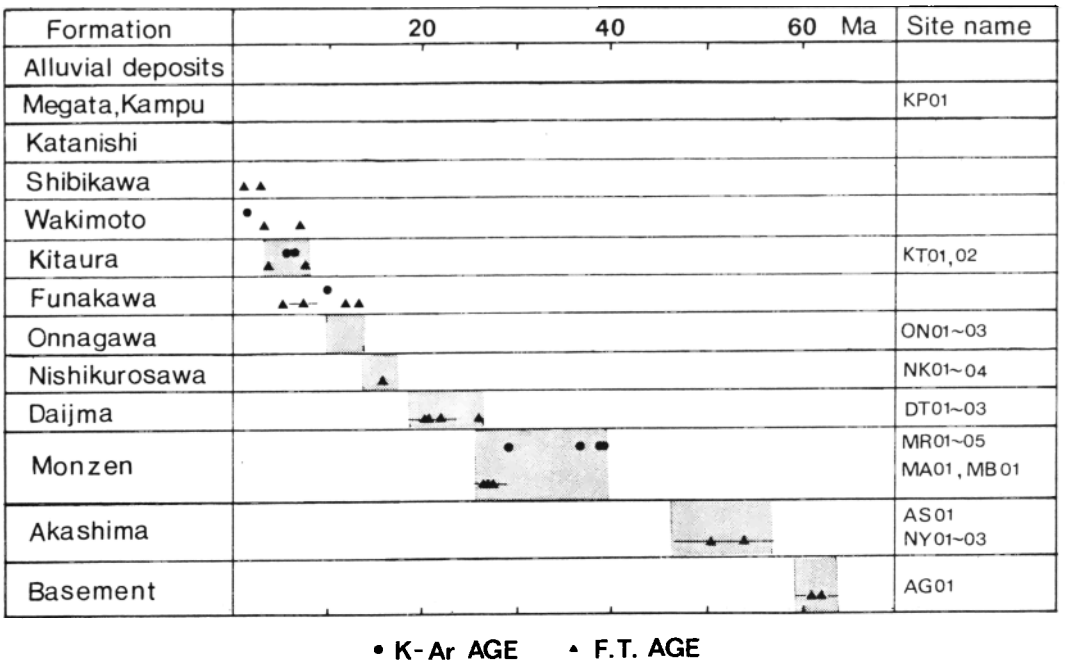


Fig. 2. Radiometric ages by K-Ar and fission track methods for the formations in the Oga Peninsula, compiled by Kaneoka [1983a]. Shaded zones indicate the ranges of the observed ages. Site names are shown in the right-hand column.

ever, recent radiometric age determinations revealed that some of the volcanic rocks erupted in the Paleogene period [Kaneoka, 1983a]. An almost complete sequence of strata with ages from early Paleogene to Quaternary is well preserved in the Oga Peninsula, on the western side of northeast Honshu at about 40° north latitude (Figure 1), making the Oga Peninsula a type locality for the Tertiary sequence in Northeast Japan.

Tertiary formations, arranged in monoclinal beds of strata, are well exposed along the west coast of the peninsula (Figure 1b). Geological units of the Oga Peninsula are roughly divided into two groups [Takayasu and Matoba, 1976]. Older formations (basement, Akashima, Monzen, and Daijima Formations) mainly consist of volcanic rocks and become younger to the south. Younger formations (Nishikurosawa, Onnagawa, Funakawa, and Kitaura Formations) are composed of sedimentary rocks and become younger to the east. The most recent volcanic rocks are Kanpuzan lava flows erupted in the Quaternary. As shown in Figure 2, over 30 age determinations made by different methods (K-Ar and fission track methods) give consistent results [Kaneoka, 1983a]. We adopted as the age of each formation the mean value of the reported ages as suggested by Kaneoka [1983a]. The ages of the younger sedimentary formations were estimated from biostratigraphic studies as summarized by Tsuchi [1981]. Previous paleomagnetic studies in the Oga Peninsula were mainly concerned with the magnetostratigraphy of the younger sedimentary rocks [Kent, 1973; Kimura, 1974; Kitazato, 1975].

In this area, a total of 26 sites were sampled, and six or more core samples were obtained at each site by using a gasoline-powered drilling apparatus. More than 150 core samples were collected in the peninsula. Sampling sites and rock types are listed in Table 1 (see also Figure 1b), with the strikes and the dips of the strata. The volcanic sequences in the peninsula are interstratified with thin sedimentary layers. The paleohorizontal plane of these rocks was estimated from the interbedded sedimentary rocks near the sampling sites. Foliations of the included pumice or ash (eutaxitic texture; [Ross and Smith, 1961]) are discernible in welded tuffs and were also used for estimating the paleohorizonal plane.

EXPERIMENTAL PROCEDURES

Specimens which are 2.5 cm in diameter and about 2.3 cm in length were cut from the collected core samples. The remanence was measured using a Schonstedt SSM-1A spinner magnetometer. Because of the weak intensities of the magnetization, tenfold to twentyfold stacking of the output signal was required for the sedimentary rocks. Progressive AF and thermal demagnetizations were carried out to obtain stable components. Heating for the thermal demagnetization was performed in a slow flow of nitrogen gas in order to prevent chemical changes of the ferromagnetic minerals in the samples. The results of the progressive demagnetization were plotted on orthogonal vector diagrams [Zijderveld, 1967], and stable and soft components of magnetization were calculated from the least squares fit of straight lines.

Figures 3a and 3b show typical examples of the orthogonal plot for the AF and the thermal demagnetization

TABLE 1. Paleomagnetic results on Seven Formations in the Oga Peninsula

Site	N	Rock Type	Bedding		Before Tilt Correction			After Tilt Correction			Pole Position		
			Strike	Dip	Inclination	Declination	α_{95}	Inclination	Declination	α_{95}	Latitute(N)	Longitude(E)	
					Basement (About 62 Ma)								
AG01	9	adamellite (far from dike)			60.5°	351.7°	4.7°				83.4°	66.1°	
AG01	4	adamellite (near dike)			-59.0°	211.1°	4.1°						
					Akashima Formation (About 52 Ma)								
AS01	4	andesite	N20°E	15°N	58.0°	348.9°	5.9°	48.4°	333.2°				
NY01	5	welded tuff			25.0°	18.9°	47.1°						
NY02	6	welded tuff	N40°W	5°S	50.3°	351.2°	9.2°	52.7°	345.7°				
NY03	7	welded tuff	N28°E	8°S	45.7°	345.4°	8.3°	50.8°	352.4°				
Mean	3	(except NY01)			51.4°	348.4°	9.9°	50.9°	343.6°	10.0°	74.3°	22.8°	
					Monzen Formation (About 35 Ma)								
MR01	9	rhyolite	N22°E	32°N	-63.3°	192.5°	6.2°	-45.9°	151.6°				
MR02	5	rhyolite	N08°W	22°W	-71.9°	146.5°	10.4°	-56.2°	112.3°				
MR03	4	rhyolite	N40°E	20°N	-71.2°	197.9°	7.7°	-58.0°	164.4°				
MR04	5	rhyolite			-59.0°	166.3°	5.1°						
MR05	5	rhyolite	N41°E	23°S	-27.6°	135.3°	7.1°	-50.5°	137.0°				
MA01	6	andesite	N70°E	28°S	-30.3°	153.5°	2.6°	-58.0°	149.3°				
MB01	4	basalt	N86°E	36°S	-17.3°	148.6°	14.2°	-47.7°	135.6°				
Mean	6	(except MR04)			-48.8°	155.4°	25.0°	-53.8°	141.7°	9.7°	59.1°	51.7°	
					Daijima Formation (About 22 Ma)								
DT01	5	welded tuff	N37°W	16°N	49.7°	295.7°	1.8°	54.5°	315.6°				
DT02	5	tuff	N40°W	15°N	50.2°	279.2°	6.4°	58.2°	297.0°				
DT03	10	tuffaceous ss	N80°E	10°S	-38.7°	124.5°	6.7°	-45.3°	117.6°				
DB01	5	basaltic tuff			74.5°	124.3°	6.6°						
DB02	5	basalt	N40°W	17°N	41.4°	302.5°	13.9°	44.3°	318.2°				
Mean	4	(except DB01)			45.4°	291.2°	11.2°	51.0°	307.4°	11.3°	47.2°	55.8°	
					Nishikurosawa Formation (About 15 Ma)								
NK01	7	siltstone	N29°E	10°S	56.8°	349.5°	8.6°	62.2°	3.9°				
NK02	7	tuffaceous ss	N29°E	10°S	55.1°	346.3°	12.6°	61.2°	359.4°				
NK03	5	siltstone	N29°E	10°S	56.5°	356.0°	10.6°	60.9°	10.9°				
NK04	7	mudstone	N55°W	7°N	-52.7°	162.8°	11.3°	-48.1°	169.2°				*
Mean	4				54.7°	350.3°	4.6°	58.4°	359.8°	9.5°	89.5°	-74.3°	
					Onnagawa Formation (About 12 Ma)								
ON01	6	fine ss	N05°E	5°N	-53.8°	179.1°	10.7°	-53.0°	172.4°				
ON02	6	fine ss	N28°E	9°S	58.9°	350.4°	11.6°	63.5°	4.6°				*
ON03	6	hard shale	N33°E	13°S	56.4°	356.5°	5.6°	62.3°	16.3°				*
Mean	3				56.4°	355.5°	5.4°	63.2°	4.5°	8.3°	84.0°	172.2°	
					Kampuzan and Kitaura Formation (0–10 Ma)								
KP01	4	andesite			72.8°	7.7°	9.7°						
KT01	4	tuffaceous ss	N30°W	25°N	51.3°	351.0°	17.7°	37.8°	12.4°				
KT02	3	tuffaceous ss	N30°W	25°N	-62.7°	148.5°	17.6°	-54.1°	188.4°				*
Mean	3				63.0°	347.1°	20.8°	54.9°	10.1°	27.3°	80.8°	-103.9°	

N denotes the number of specimens used to compute the site-mean direction. Inclination, Declination, and α_{95} are inclination, declination, and semi-angle of the cone of 95% confidence in degrees, respectively. Latitude and Longitude are latitude and longitude of pole positions in degrees corresponding to the formation mean directions, respectively. In some sites, field directions after tilting correction are not present because of uncertainity as to the paleohorizontal plane (AG01, NY01, MR04, DB01, and KP01).

* Both normal and reverse magnetized samples, which were almost antipodal to one another, were obtainedfrom the site.

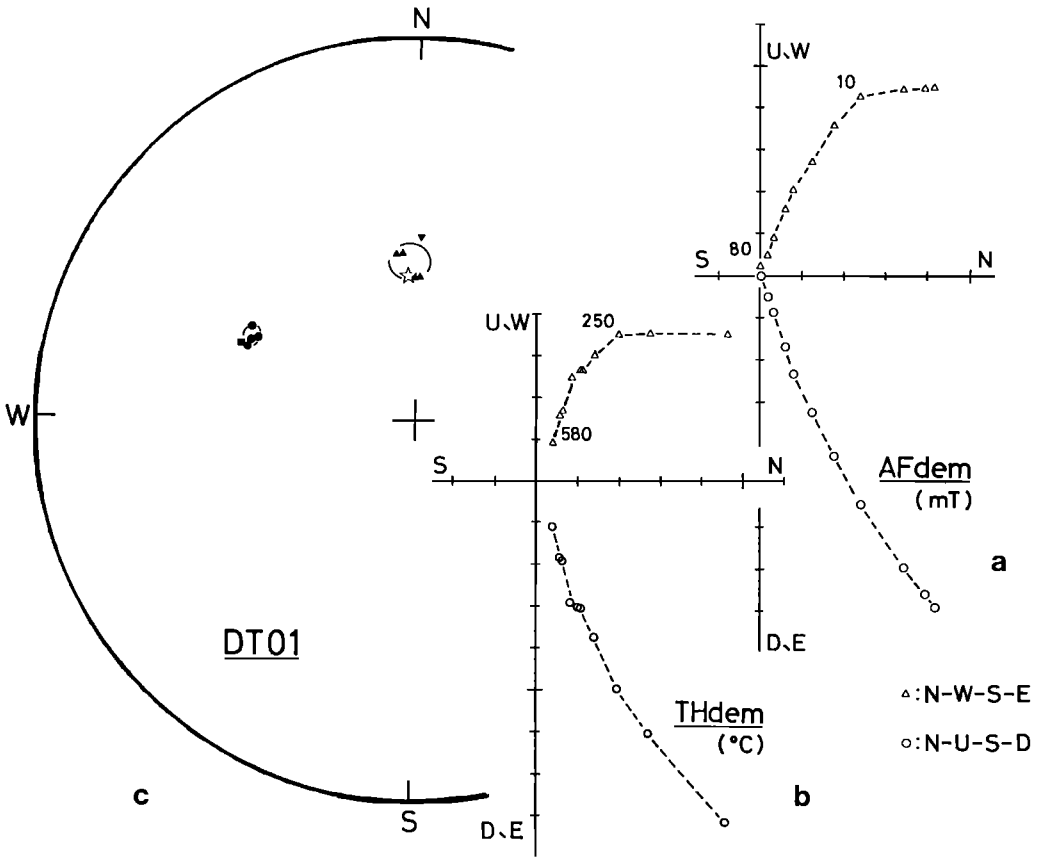


Fig. 3. Typical examples of the orthogonal vector diagram for the (a) stepwise AF and (b) thermal demagnetization curves. The units are milliteslas in AF demagnetization and degrees Celsius in thermal cleaning. (c) Equal-area projection for the stable and secondary components. Symbols for the equal-area projection are as follows: circle, stable components obtained by AF demagnetization; square, stable components by thermal cleaning; upper triangle, soft components by AF demagnetization; lower triangle, secondary components by thermal cleaning. Open star in Figure 3c denotes the axial geocentric dipole field at this site.

curves, respectively. These figures indicate that the lower coercivity and the lower blocking temperature components can be removed by the first few steps and that there is no significant difference between the two demagnetization processes. As shown in Figure 3c, the soft components are close to the direction of the geocentric axial dipole field. In the present study, most of the stable components of magnetization were determined from stepwise AF demagnetization, whereas one or two specimens from each site were subjected to thermal demagnetization to evaluate the validity of the stable direction.

RESULTS

For the compilation of the results, we divide the present samples into seven geological units: basement, Akashima, Monzen, Daijima, Nishikurosawa, Onnagawa, and Kitaura-Kanpuzan Formations. The site names, assigned ages, number of samples, bedding, and estimated site-mean directions before and after tilting correction are summarized in Table 1. In the following we examine the validity of the data for the ancient field direction.

The upper three units (Onnagawa, Nishikurosawa, and Kitaura Formations) consist mainly of sedimentary rocks and yielded normally and reversely magnetized samples. This assures the averaging of the secular variation. The mean directions for these units are close to the present north-south direction.

The Daijima Formation shows the largest westward shift among the present units. The sampled rocks consist of basaltic and tuffaceous rocks. The original orientation of the flow could not be determined at site DB01. Hence the calculation of the formation mean direction was made excluding this site. The antipodal nature of the normal and the reversed directions and the large difference between the present field direction and the in situ site-mean directions indicate the original nature of the magnetization.

The Monzen Formation has the largest exposure and thickness in the western part of the Oga Peninsula, where seven sites were sampled for the present study. All the stable directions obtained for these samples are reversely

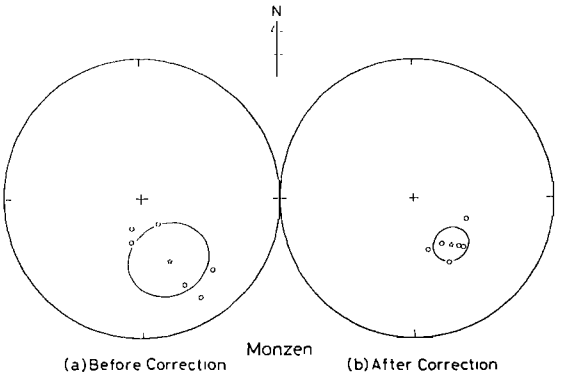


Fig. 4. Site-mean directions (a) before and (b) after bedding correction for six sites of the Monzen Formation. Solid and open symbols correspond to the directions in the lower and upper hemispheres, respectively. Stars indicate the mean directions, and circles around the stars correspond to 95% confidence limits.

magnetized and indicate an anticlockwise rotation of the area. Since most of the strata in the peninsula are monoclinal except the Monzen Formation, they are not suitable for the fold test. In the Monzen Formation, two sides of a syncline could be sampled, where three sites (MR01-03) are from the southern limb and the others (MR05, MA01 and MB01) are from the northern limb. Although the in situ direction of site MR04 is close to the other sites, the data are excluded because there are no interbedded sediments to indicate the original horizontal datum. Figure 4 shows the site-mean directions before and after horizontality correction for the remaining six sites. The effectiveness of the correction as evident from Figure 4 and the statistical test proposed by McFadden and Jones [1981] indicate that the remanences were acquired before the folding.

Four sites are sampled in the Akashima Formation. Site NY01 is situated very close to the boundary between the Akashima and the Monzen Formations, and the rocks appear to have been severely deformed, as indicated by microcracks and subdivision into many small blocks. Figure 5 shows the stable directions from the four sites in the Akashima formation. The directions from three sites (AS01, NY02 and NY03) have slightly westward declinations, whereas the directions from site NY01 show a wide scatter. For the calculation of the formation mean direction, we used the three sites excluding NY01. Since the mean direction is close to the present field direction, contamination by VRM (Viscous Remanent Magnetization) due to the present field might be possible and should be examined. For this purpose, we used the data for site NY01 as a test. The similarity of the AF and the thermal demagnetization characteristics of the samples from site NY01 to those of the other three sites supports the same origin of these remanences. However, the stable directions from site NY01 are scattered as shown in Figure 5, suggesting the effect of later deformation. This scatter indicates that the remanences were acquired before the deformation. Assuming that the timing of the deformation was pre-Tertiary, the remanences were not of VRM origin. Also, the magnetic characteristics of the rocks from the Akashima Formation are similar to those of the other volcanic rocks from the Monzen and the Daijima Formations. It is rather unreasonable to assume that only the Akashima Formation was severely affected by VRM. VRM components which can be clearly identified from the other volcanic rocks (the Monzen and the Daijima Formations) have much lower coercivity and the lower blocking temperature.

Otofuji et al. [1985] also sampled the Akashima Formation at four sites. Their in situ directions are similar to the present results. However, after horizontality correction, their data show a wide scatter, suggesting high angles of the estimated dip in some of their sites. On the basis of this result, they concluded that the remanences are VRM. Although the strikes and dips of strata for their sampling sites are not listed in their paper, we can estimate the values from their magnetic directions before and after the correction. The calculated strikes and dips are not concordant with the overall gentle and monoclinical nature of the Akashima Formation as shown in the geological map of this area [Huzioka, 1959]. Hence we may conclude that their sites are in a highly deformed region, where the estimate of the bedding for the volcanic rocks might be very difficult.

Basement rocks in the Oga Peninsula consist of hornblende-biotite adamellite. The rocks were sampled at one site (AG01), where they are cut by a basaltic dike about 1.7 m thick. As reported by Tosha and Hamano [1986], many dikes are observed along the west coast of the Oga Peninsula, mostly among the Monzen and the Akashima Formations. These dikes have northeast-southwest strikes and are considered to have intruded during the Nishikurosawa or the Onnagawa stages. The northward directions of the remanence exhibited by these dikes are consistent with the assigned age [Tosha and Hamano, 1986]. The sampled dike intruding the basement is con-

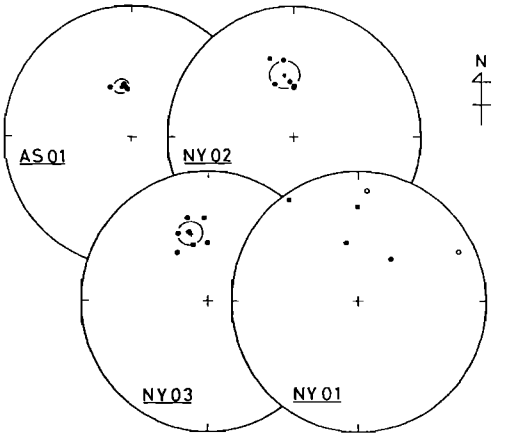


Fig. 5. Stable components of the remanences for the samples from the four sites of the Akashima Formation before tilting correction.

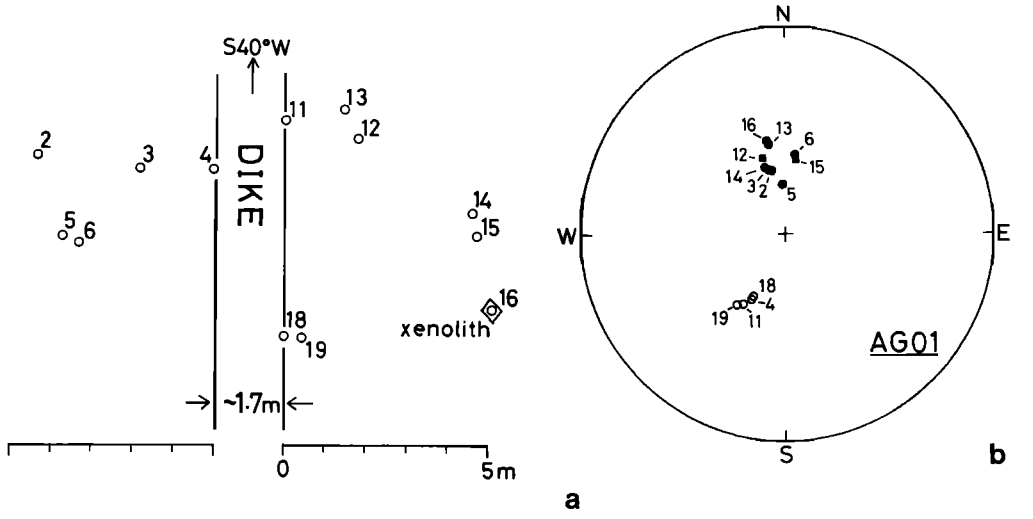


Fig. 6. (a) Sample locations at site AG01. The dike intrudes the basement and strikes S40°W. (b) Stable components of the remanence plotted on an equal-area projection. Solid and open symbols correspond to the lower and the upper hemispheres, respectively. Circles and squares indicate the stable direction obtained by the AF and the thermal demagnetizations, respectively.

sidered to be one of these dikes from its strike, rock type, and direction of stable remanence.

For the purpose of the baked contact test, samples of the basement were obtained at distances of 0.5 m to 5 m from the dike, and the dike itself was sampled. The relative position of the samples is shown in Figure 6a, and the stable directions for these samples are shown in Figure 6b. The four samples obtained at the contact of the dike give reversed directions which are close to the mean direction of the samples from within (Dec=199°, Inc=−39°, and $\alpha_{95} = 19°$). On the other hand, other samples show normal directions. We adopt the normal direction as the paleomagnetic field direction for the basement. The direction is close to that of the Akashima Formation and the present geomagnetic field. Hence the effect of the VRM component should also be examined. Thermomagnetic analysis for the rocks close to and far from the contact indicates that they have a Curie temperature of about 600°C, indicating magnetite as the main magnetic phase in these rocks. It is also inferred that heating of the basement rocks due to the dike intrusion did not severely affect the magnetic minerals in the rocks. The VRM component in the reversely magnetized samples can be demagnetized by an AF field of 5 mT or a temperature of 200°C. Assuming that the coercivity and the blocking temperature spectra of the magnetite in these rocks were also unchanged by the heating, this suggests that the stable component of the basement is not a VRM.

Although the possibility of VRM is considerably diminished by the above test, the data should be treated with some caution, since the paleohorizontal plane of the basement rocks cannot be estimated, and the heating effect due to the Akashima lava flows cannot be estimated. However, it is inferred from the above discussion that the remanence was acquired not later than the Akashima age (about 50 Ma). We tentatively assume that the remanence represents the field direction at the age of 62 Ma.

Site mean directions for the seven formations are shown in Figure 7, and the temporal variations of the inclination and the declination of the formation mean directions

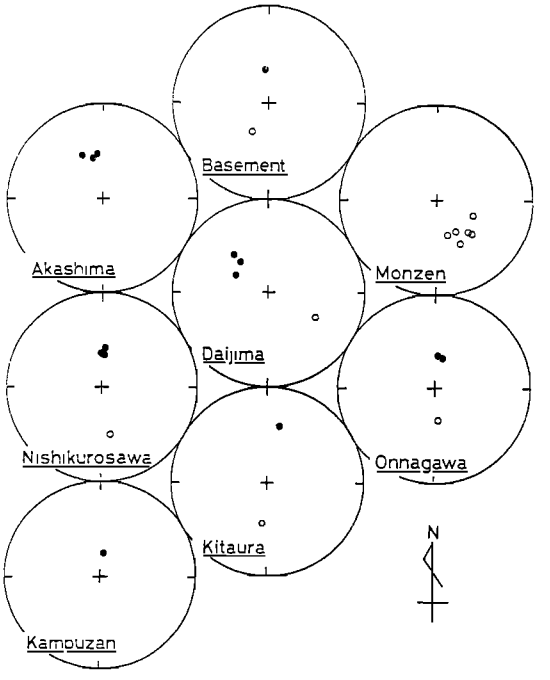


Fig. 7. Equal-area projection of the tilt-corrected site-mean directions for the eight formations in the Oga Peninsula.

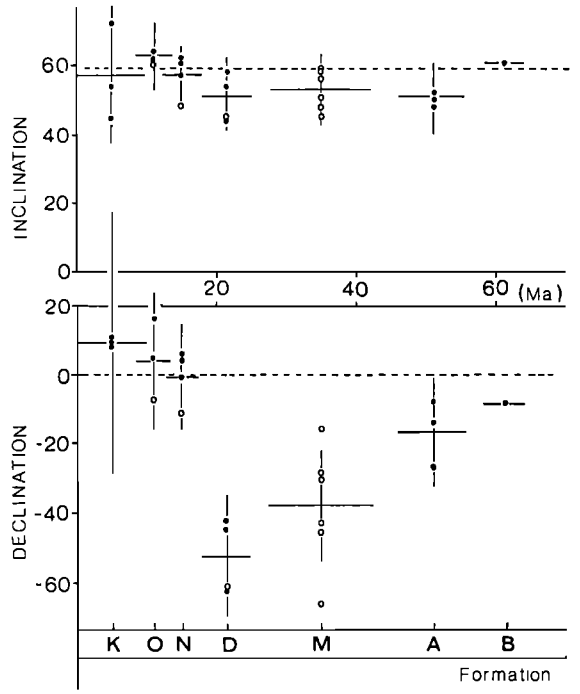


Fig. 8. Variations of inclination and declination with age. Dotted lines indicate the inclination (59°) and the declination (0°) expected for the geocentric axial dipole field. The horizontal and the vertical bars for each data set indicate the possible error expected for the age and the paleodirection, respectively.

are plotted in Figure 8. All the data except for the basement are the directions after the horizontality correction. These figures indicate a very small change of the inclination during the entire Tertiary and Quaternary periods. On the other hand, the declination gradually changes to a westward direction until about 20 Ma, and then abruptly changes to a northerly direction.

DISCUSSION

In order to discuss the tectonic movements of geological blocks from the paleomagnetic results, we need a reference pole from a stable region. Previous paleomagnetic studies on the tectonic movements of the Japanese Islands have attempted to relate the movement derived from the paleomagnetic study to the opening of the Sea of Japan. For this purpose, we need a reference pole from the Asian continent. However, very few paleomagnetic results are available in this area for Tertiary time (for example, see Khramov et al. [1981]). In the previous studies, therefore, the poles for the western Eurasian plate [Irving, 1977] were used as reference poles. However, since Eastern Asia seems to have been deformed in the Cenozoic by the collision of the Indian plate [e.g., Tapponnier et al., 1982], it is not appropriate to use the Eurasian poles for the discussion of the tectonic movements within Eastern Asia. Hence we use Korean data to investigate the motion of Northeast Japan relative to Eastern Asia.

The APWP (Apparent Polar Wander Path) for Northeast Japan obtained by the present study and that for the Korean Peninsula from Kang [1964], which was compiled by Sasajima [1981], are plotted in Figure 9. The oldest pole of about 100–110 Ma for Northeast Japan was obtained by Tosha [1983] from the Miyako group in the northern Kitakami area (see, Figure 1a). Although the ages for the Korean APWP are not precisely determined, the two APWPs resemble each other for the period ranging from the Early Cretaceous to the Eocene. As shown in Figure 10a, these parts of the respective APWPs approximately coincide when we rotate Northeast Japan clockwise by 22.8° around a pole at the latitude of 58.1°N and the longitude of 148.9°E. The rotation moves Northeast Japan to the position indicated in Figure 10b if the Korean Peninsula is fixed to its present position. Therefore we can conclude that Northeast Japan and the eastern margin of the Asian continent belonged to one tectonic block during the Cretaceous and the Eocene periods, and then relative movement between the two occurred in the Miocene. The timing of the relative movement is not clear from the fitting of the APWPs because the paleomagnetic poles have their own statistical errors and the ages corresponding to the poles are not precisely determined. Strictly speaking, the temporal declination change in Northeast Japan does not mean the relative movement between the two tectonic blocks but indicates the movement relative to the north pole. However, as shown in Figure 8, the gradual westward change in declination suddenly turns to the north between 22 Ma and 15 Ma. Figure 8 suggests that the movement occurred at that period. The fitting of the APWPs indicates that the amount of counterclockwise rota-

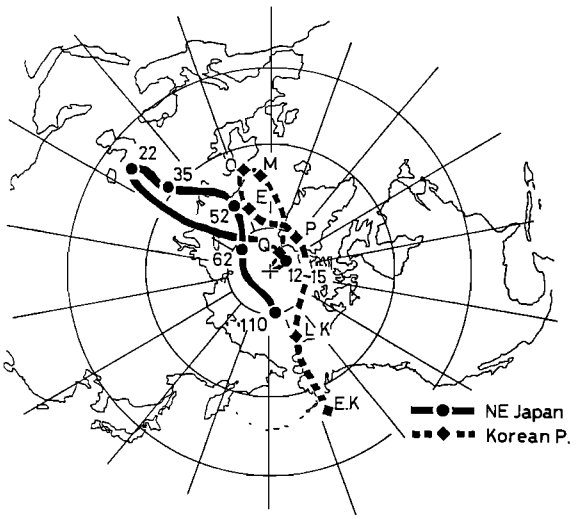


Fig. 9. Apparent polar wander paths for Northeast Japan and the Korean Peninsula from the Early Cretaceous to the present. Numbers and captial letters correspond to the assigned ages in Ma for Northeast Japan, and in geological periods and epocks for the Korean Peninsula, respectively.

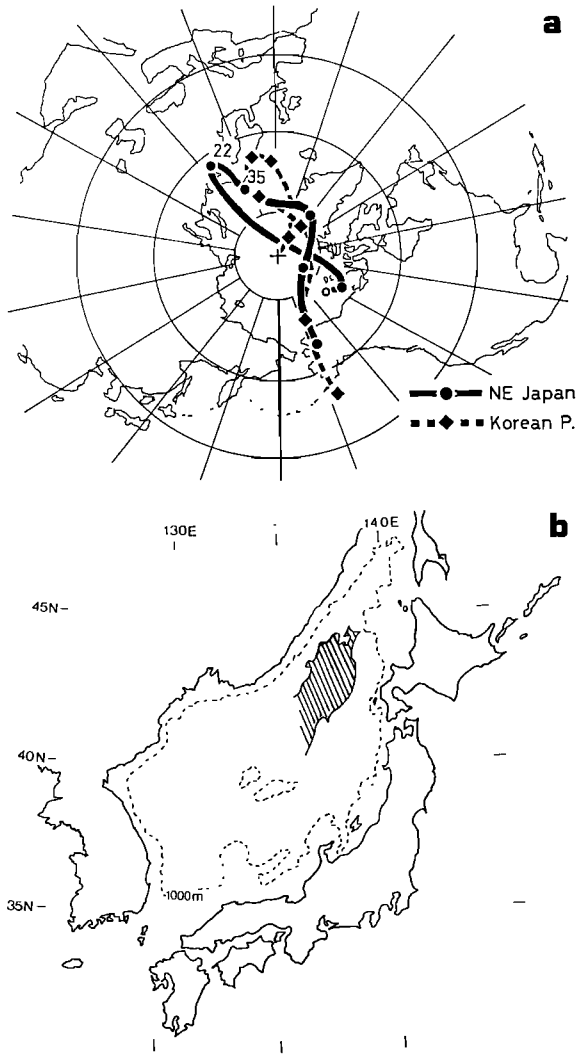


Fig. 10. (a) Apparent polar wander paths for the Korean Peninsula and Northeast Japan, assuming the position of Northeast Japan relative to the Korean Peninsula as indicated in Figure 10b. (b) The hatched position of Northeast Japan is obtained by a rotation of 22.8° around a pole at 58°N, 149°E.

tion of Northeast Japan relative to the eastern margin of Asia is about 20°, which is much smaller than the amount of rotation relative to western Europe (see, for example Otofuji et al. [1985]). This may suggest that at that time the eastern margin of Asia also deformed with respect to the Eurasia block.

It is not unreasonable to relate the rotation of Northeast Japan to the opening of the Sea of Japan, although, in general, the opening can occur without rotating the Japanese Islands. However, the following pieces of evidence are consistent with the assumption that the rotation of Northeast Japan accompanied the separation of the block from the Asian continent. Togashi [1983] measured chemical compositions of Neogene volcanic rocks erupted in Northeast Japan and concluded that the magma for these rocks with ages about 25 to 22 Ma was generated under almost the same conditions as those of the present continental arc type activities, while the magmatism became the island arc type after 22 Ma. By compiling the initial ratios of $^{87}Sr/^{86}Sr$ of volcanic rocks in Northeast Japan, Kaneoka [1983b] pointed out that the rocks erupted before 16 Ma have higher initial ratios than those of recent rocks. The higher initial ratios observed in early Miocene volcanic rocks seem to suggest the existence of an underlying continental lithosphere before 16 Ma.

Celaya and McCabe [1987] proposed a two-stage opening of the Sea of Japan by using Tosha's [1983] data, and they attributed the first gradual change of the declination before the Miocene to the first clockwise rotation of Japan and the abrupt change at about 20 Ma to the second counterclockwise rotation of Northeast Japan. However, a declination change in paleomagnetism means no more than a relative movement to the north magnetic pole. The relative movement between the Japanese Islands and Eastern Asia must be discussed on the basis of two paleomagnetic data sets in these regions. Since the first gradual change can also be observed in the Korean data and even in the data from western Europe [Irving, 1977], it is not reasonable to relate the first change to the opening of the Sea of Japan.

CONCLUSION

Based on paleomagnetic study, we conclude that Northeast Japan was situated along the east coast of the Asian continent since Early Cretaceous and rotated apart during the period between 22 and 15 Ma. The amount of the rotation is estimated to be roughly 20° counterclockwise.

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